GENBE301 – APPLY BASICS OF ELECTRICITY

DATE: …/…/2025

PERIOD: ……..

TECHNICAL SECONDARY SCHOOLS (TSS)

END OF TERM I EXAMINATIONS: 2025-2026

QUESTION PAPER

TRADE NAME: NIT&ETE

RQF LEVEL: 3

MODULE TITLE: APPLY Basics of electricity

MARKS: **……/100**

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed Sections (A, B, C). Follow the instructions given below, and answer the indicated questions in each section for a total of 100marks

Section **A**: 26 questions, all are **Compulsory**. **55 marks**
Section **B**: Five 5 questions, choose and answer any three (3). **30 marks**
Section **C**: Two 2 questions, choose and answer any one (1) **15 marks**

Allowed Materials: Blue pen, Ruler, Calculator, Pencil, (Fitting with the exam paper)

Section **A**: 26 questions, all are **Compulsory**. **55 marks**

MULTIPLE-CHOICE QUESTIONS (20 marks)

1. Alternating current (AC) is defined as:
A. Current that flows in one direction
B. Current that flows in two directions
C. Current with zero resistance
D. Current used only in batteries

2. The unit of electric current is:
A. Volt
B. Ohm
C. Ampere
D. Watt

3. Voltage is defined as:
A. The flow of electrons
B. The electrical pressure that moves electrons
C. The resistance to electron flow
D. The energy consumed per hour

4. Which instrument is connected in series?
A. Voltmeter
B. Ohmmeter
C. Wattmeter
D. Ammeter

5. Resistance is measured in:
A. Volts
B. Ohms
C. Amperes
D. Watts

6. Conductance is the reciprocal of:
A. Voltage
B. Current
C. Resistance
D. Power

7. Ohm's law states that:
A. I = V + R
B. V = I/R

C. $I = V/R$
D. $P = V \times I$

8. In a series circuit:
A. Voltage is the same across each resistor
B. Current is the same through all resistors
C. Total resistance decreases
D. Conductance is additive

9. In a parallel circuit:
A. Voltage is different across each branch
B. Resistance is additive
C. Current divides among the branches
D. Same current flows everywhere

10. Kirchhoff's Current Law states:
A. Sum of voltages in a loop equals zero
B. Total current entering a junction equals total leaving
C. Total resistance equals sum of individual resistances
D. Voltage is directly proportional to current

TRUE OR FALSE QUESTIONS (20 marks)

1. Electron flow is from negative to positive terminal.

2. Voltage is also called electrical pressure.

3. A voltmeter is connected in series in a circuit.

4. Power is given by $P = V \times I$.

5. Conductors have high resistance.

6. Semiconductors have conductivity between conductors and insulators.

7. In a series circuit, voltage remains the same across each resistor.

8. In a parallel circuit, voltage is the same across all branches.

9. According to Kirchhoff's Voltage Law, the sum of voltage drops equals the total EMF.

10. The unit of conductance is Siemens (S).

MATCHING QUESTIONS (5 marks)

Match Column A with Column B

Column A

1. Ammeter
2. Ohmmeter
3. Conductor
4. Insulator
5. Semiconductor

Column B

A. Used to measure resistance
B. Allows current to flow easily
C. Material with medium conductivity
D. Does not allow current to flow
E. Measures current in a circuit

FILL-IN-THE-BLANK QUESTIONS (10 marks)

1. The unit of voltage is the _____.

2. Resistance opposes the _____ of current.

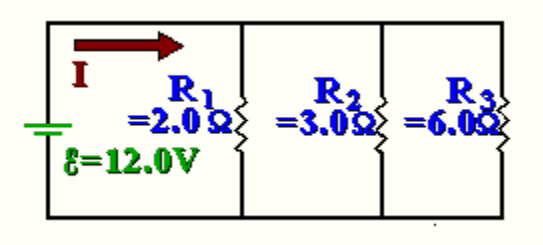
3. The formula for electrical power is P = _____ × I.

4. In a parallel circuit, the _____ is the same across all branches.

5. According to Ohm's law, I = _____ /

Section **B**: Five 5 questions, choose and answer any three (3). **30 marks**

1. In the following schematic diagram, find the total current (I).

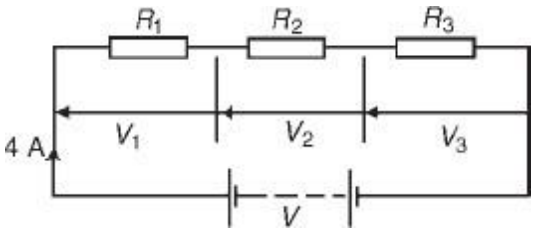


2. A 100 W electric light bulb is connected to a 250 V supply. Determine (a) the current flowing in the bulb, and (b) the resistance of the bulb
3. Explain The main characteristics of a series circuit

4. Using Kirchhoff's Voltage Law, describe how voltage drops behave in a closed loop circuit
5. Describe the behavior of current and voltage in a parallel circuit and give a practical example

Section **C**: Two 2 questions, choose and answer any one (1) **15 marks**

1. For the circuit shown in Figure below, determine
(a) The battery voltage V,
(b) The total resistance of the circuit, and
(c) The values of resistance of resistors R1, R2 and R3, given that the p.d.'s across R1, R2 and R3 are 5 V, 2 V and 6 V respectively. /5marks



2. For the circuit shown in Figure below, determine
(a) The reading on the ammeter, and
(b) The value of resistor R2 /4marks

